

epoc Blood Analysis System Product Specifications

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System Description

Patient-side blood gas analyzer

System Menu

pH	Na ⁺	Hct	BUN
pCO ₂	K ⁺	Glu	Urea
pO ₂	Ca ⁺⁺	Lac	
TCO ₂	Cl ⁻	Crea	

Measured Parameters

Parameter	Unit of Measure	Measurement Range
pH	pH units	6.5–8.0
pCO ₂	mmHg kPa	5–250 0.7–33.3
pO ₂	mmHg kPa	5–750 0.7–100
TCO ₂	mmol/L mEq/L	5–50 5–50
Na ⁺	mmol/L mEq/L	85–180
K ⁺	mmol/L mEq/L	1.5–12.0
Ca ⁺⁺	mmol/L mg/dL mEq/L	0.25–4.00 1.0–16.0 0.5–8.0
Cl ⁻	mmol/L mEq/L	65–140
Hct	% PCV L/L	10–75 0.10–0.75

Measured Parameters continued

Parameter	Unit of Measure	Measurement Range
Glu	mmol/L mg/dL g/L	1.1–38.5 20–700 0.20–7.00
Lac	mmol/L mg/dL g/L	0.30–20.00 2.7–180.2 0.03–1.80
Crea	mg/dL μmol/L	0.30–15.00 27–1326
BUN	mg/dL	3–120
Urea	mmol/L mg/dL g/L	1.1–42.8 7–257 0.07–2.57

Institutions should establish and set their own normal range values.

Calculated Parameters

Parameter	Unit of Measure	Measurement Range
cHgb	g/dL mmol/L g/L	3.3–25 2.0–15.5 33–250
cHCO ₃ ⁻	mmol/L mEq/L	1–85
cTCO ₂	mmol/L mEq/L	5–50
BE(ecf)	mmol/L mEq/L	-30–+30
BE(b)	mmol/L mEq/L	-30–+30

Calculated Parameters continued

Parameter	Unit of Measure	Measurement Range
cSO ₂	%	0–100
GFRmdr*†	mL/min/ 1.73m ²	2–60 or >60
GFRmdr-a*†	mL/min/ 1.73m ²	2–60 or >60
GFRckd [‡]	mL/min/ 1.73m ²	1–225
GFRckd-a [‡]	mL/min/ 1.73m ²	1–225
GFRswz [§]	mL/min/ 1.73m ²	1–275
AGap	mmol/L mEq/L	-14–+95
AGapK	mmol/L mEq/L	-10–+99
BUN/Crea	mg/mg	0.2–400.0
Urea/Crea	mmol/ mmol mg/mg	0.8–1615.4 0.4–856.8
A	mmHg kPa	5–800 0.67–106.64
A-a	mmHg kPa	1–800 0.13–106.64
a/A	% fraction	0–100 0–1

epoc® Blood Analysis System Product Specifications (continued)

Sample Types

Whole blood: arterial, venous, capillary

Sample Size

At least 92 µL

Time to Result

Less than 1 minute

Calibration

Automatically performed prior to every test

Quality Control

Eurotrol GAS-ISE Metabolites (with BUN)
Eurotrol Calibration Verification Fluids (with BUN)
Eurotrol Hct Control
Eurotrol Hematocrit Verification Fluid

Integrated Bar-code Scanner

For Patient ID, Operator ID, and ampuled QC;
1D and 2D bar-code symbologies

External Interfaces

HL7

Power Requirements

100–240 VAC; 50–60 Hz

Operation on Battery

At least 50 test cards

Safety

IEC 61010-1, IEC 61010-2-81,
IEC 61010-2-101

EMC

IEC 61326-1, IEC 61326-2-6

Data Security

Patient Data encryption
Encrypted end-to-end data transmission
No hard-coded password
No user access to operating system

Hardware

Operating System	Android 9.0
Processor	MT6762 8*Cortex-4*A53 2.0GHz/4*A53 1.5GHz
Memory	2 GB RAM; 16 GB ROM

BLUETOOTH

BT4.2(BLE HS compliant) BT Smart Ready

BLUETOOTH Security

128-bit encryption with PIN authentication

WIFI

MediaTek 6631 chipset
802.11 a/b/g/n/ac/r
2.4GHz +5GHz**

WIFI Security

WEP, WPA/WPA2 Personal (Pre-Shared Key)
EAP (802.1x): TLS, TTLS (MS-CHAP), TTLS (MS-CHAP v2), TTLS (CHAP),
TTLS-MD5, TTLS-PAP, PEAP-TLS, PEAP (MS-CHAP v2), LEAP,
FAST WAPI PSK, WAPI Certificate

Data Capacity

Patient Test & QA	2499
Operators	4000

Test Card

Storage	15–30°C
Shelf Life	Up to 5 months
Size	3.39 in (L) × 2.13 in (W) × 0.06 in (H) 86 mm (L) × 54 mm (W) × 1.4 mm (H)

System Dimensions

	Host	Reader
Height (nominal)	160 mm [6.30 in]	215 mm [8.46 in]
Width (nominal)	78 mm [3.07 in]	85 mm [3.35 in]
Thickness (nominal)	16 mm [0.63 in] (22 mm [0.87 in] at scanner)	50 mm [2 in]
Weight (nominal)	250 g [8.82 oz] (with battery)	<500g [<1.1 lb]
Display	5.0" diagonal, HD 1280 × 720 pixels, touchscreen	—

Environmental Requirements

Operating Temperature	-20° C to 55° C [-4° F to 130° F] (Host)	15°C–30°C (Reader and Test Card)
Humidity	Up to 85% relative humidity, non-condensing	
Barometric Pressure	400–825 mmHg (53.33–110 kPa)	

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Product availability may vary from country to country and is subject to varying regulatory requirements. Please contact your local representative for availability.

**Values >60 will be reported as
>60 mL/min/1.73 m²
†IDMS-traceable MDRD type
‡CKD-EPI equation
§Bedside Schwartz equation
**5GHz Wi-Fi Channel might not
be available in some countries*

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